

Qatar-1b

R-2595

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140922 · created 2026-08-27T23:49:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456922.81436 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.048
Transit depth (Rp/Rs)^2	3.69 +/- 1.86 [%]
Orbital Inclination (inc)	82.27 +/- 2.34
Airmass coefficient 1 (a1)	1.014 +/- 0.052
Airmass coefficient 2 (a2)	-0.007 +/- 0.029
Scatter in the residuals of the lightcurve fit is	5.06 %
Best Comparison Star	#2 - [221, 405]
Optimal Aperture	2.76
Optimal Annulus	6.55
Transit Duration (day)	0.055 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.048 vs 0.1463 (deviation 0.69σ)
Duration fit vs published	0.055 d vs 0.0692 d (deviation 0.43σ)
Mid-time O-C	-0.9 min
Depth SNR	2.9
Residual scatter	5.06 %

Light curve

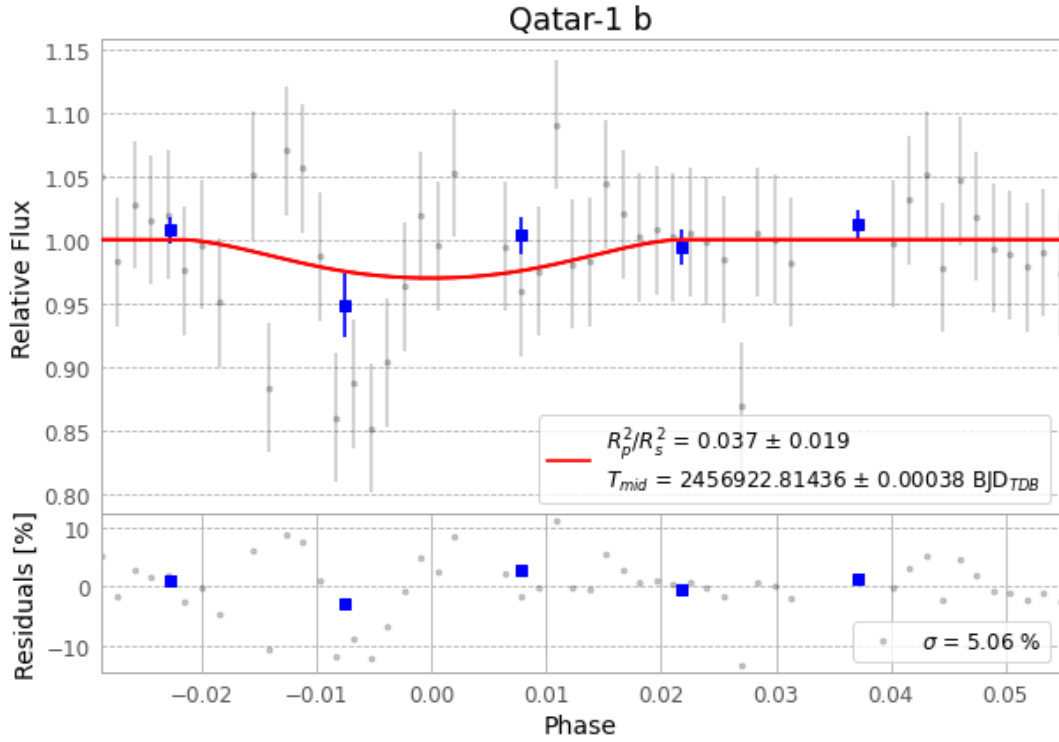


Figure 1 — FinalLightCurve_Qatar-1 b_2014-09-21.png (SHA-256 b7ca0bc7d13d4ec3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4ba503f41437f14e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), Qatar-1140922063012.FITS → Qatar-1140922092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140922.log (1a66b9e4f717...), FinalLightCurve_Qatar-1 b_2014-09-21.png (b7ca0bc7d13d...), FinalLightCurve_Qatar-1 b_2014-09-21.csv (ac62453230c8...)

Compute resources

Wall time 184.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:49Z.